Review of Cybersecurity Breaches Worldwide

Kailyn Philpot

Math 014-01: Intro. to Data Science

Dr. Meenakshi Nerolu

April 29th, 2025

The Rising Threat of Cyberattacks in a Digitally Connected World

Cybersecurity is a major global issue. Cyberattacks today don't just target individuals — they impact governments, banks, hospitals, and entire countries. A real-world example is the hacking collective known as Anonymous. Over the last decade, Anonymous has launched cyberattacks against government websites, multinational corporations, and financial institutions as a form of digital protest or political retaliation. While often described as a vigilante group, their actions demonstrate a deeper, more troubling reality: the tools and technologies they use are highly advanced and widely accessible — capable of bypassing major defenses and paralyzing critical systems.

Their efforts, although sometimes aligned with activist causes, are proof that this level of cyberpower exists outside official channels — and that such power, in the wrong hands, could lead to devastating consequences. The rise of cyber collectives like Anonymous is not only a sign of growing digital resistance, but also a warning: the potential for large-scale technological warfare is no longer theoretical. As global conflicts shift into the digital realm, cybersecurity has become not just a technical concern, but a matter of national and international security.

Objective

The goal of this project was to explore patterns in global cybersecurity incidents, focusing on attack types, financial losses, and trends over time. Studying incidents from 2015 to 2024 shows how cybersecurity threats have evolved recently and helps predict and mitigate future risks as technology and attack methods continue to change.

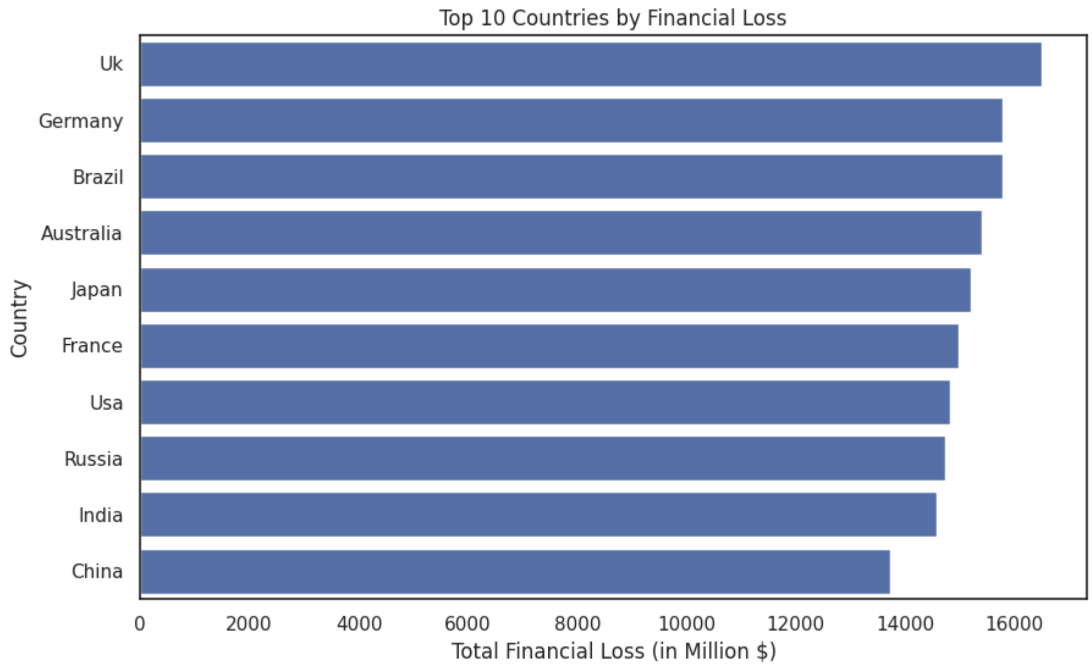
Method

This project began with a raw dataset of the global cybersecurity incidents from the last decade (2015 to 2024). The dataset included columns such as attack type, target industry, country, financial loss (in million USD), number of affected users, resolution time (in hours), and defense mechanisms used. Before any analysis could be done, the dataset required several cleaning and transformation steps to ensure accuracy and consistency.

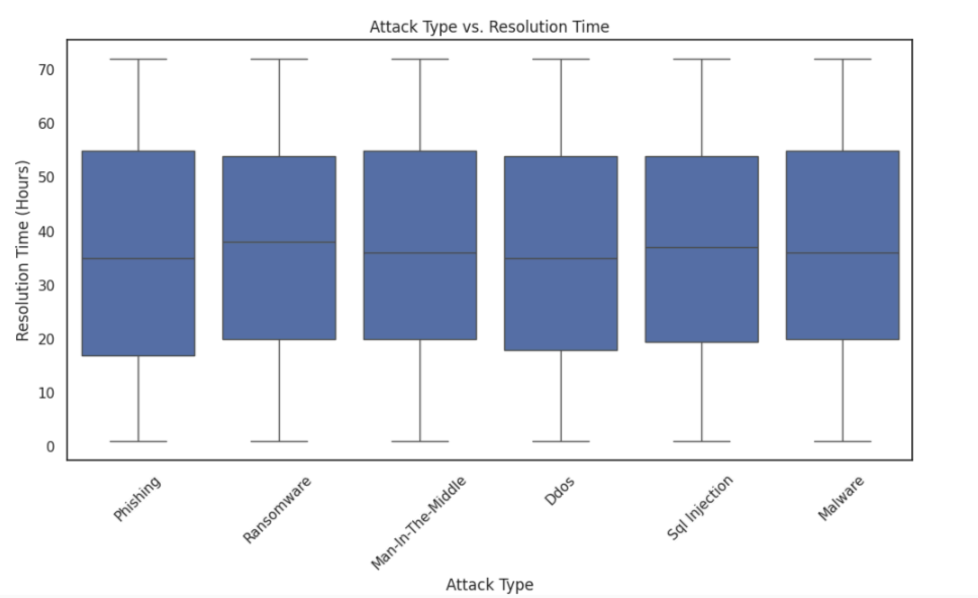
First, the data was examined for null values, duplicate records, and inconsistencies in text formatting. Any duplicate entries were removed, and missing values were identified and either filled appropriately or left out of specific visualizations to preserve data integrity. Text formatting was standardized—columns like “Attack Type” and “Country” were capitalized consistently to avoid misgrouping during analysis.

The ‘Year’ column was converted to a categorical data type since years function more like labels than continuous numerical values. This change improved grouping operations and visualization accuracy. In addition, derived columns were created to enrich the analysis. One such column, Loss per User, was generated by dividing the financial loss by the number of affected users. This metric helped measure the intensity of each attack on a per-person basis. Another column, Year Interval, grouped years into broader periods (e.g., 2015–2019 and 2020–2024), which helped identify larger trends over time rather than focusing only on individual years. These cleaning and preparation steps ensured the dataset was ready for meaningful analysis. By transforming the raw data into a cleaner and more structured format, I was able to uncover clearer patterns, compare variables accurately, and draw more reliable conclusions from the visualizations.

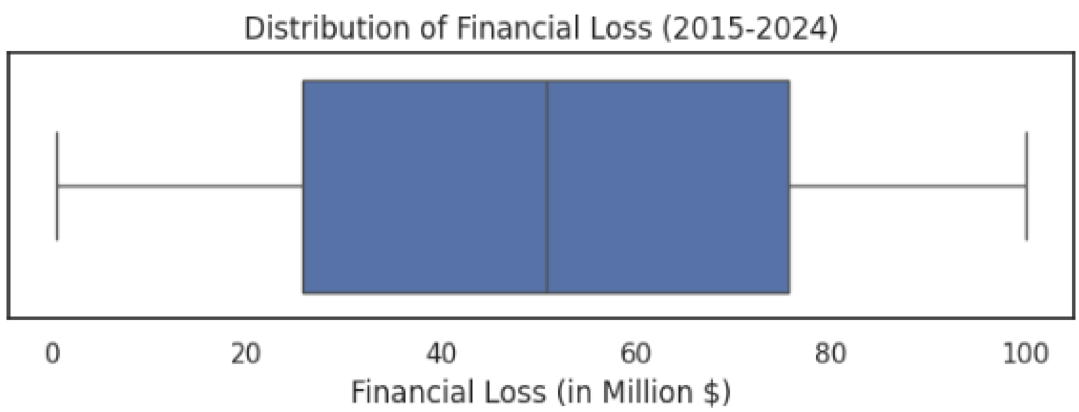
Data Visualizations & Interpretation



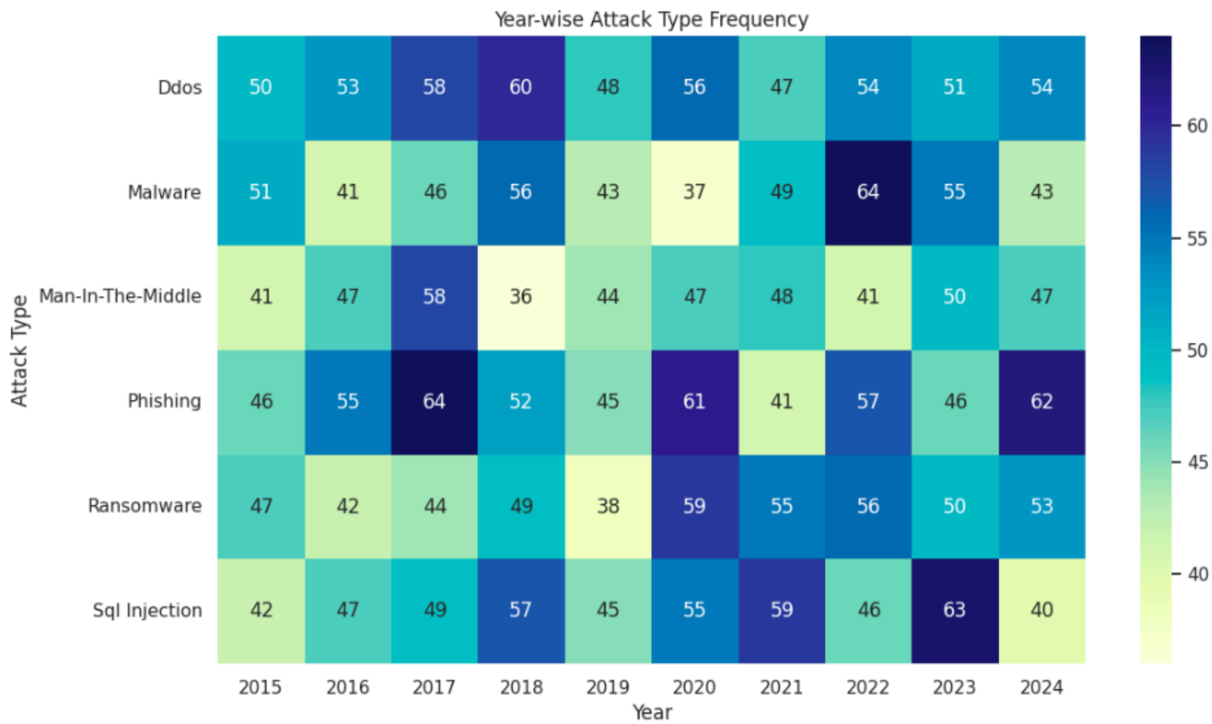
The highest total financial losses from cybersecurity incidents between 2015 and 2024. The United Kingdom ranks first, followed by Germany and Brazil, which is somewhat surprising given the expectation that the United States or China would typically dominate. This suggests that cybercriminals are increasingly targeting a broader range of countries, not just the largest global economies. It also highlights how developed digital infrastructures — like those in Europe and South America — are vulnerable to costly attacks if defenses are not strong enough.



This boxplot compares how long different types of cybersecurity attacks took to resolve, measured in hours. Each box shows the distribution of resolution times for each attack type, and the median line inside each box represents the typical resolution time. What's noticeable is that the median resolution times across all attack types are very similar, and the interquartile ranges are also fairly consistent. This means that regardless of the attack type — whether phishing, ransomware, insider threats, or malware — most incidents were resolved within roughly the same time frame. The data suggests that organizational response processes may be standardized, handling different types of attacks with similar efficiency.



This boxplot shows the distribution of financial losses from cybersecurity incidents between 2015 and 2024. In a boxplot, the middle box represents the range between the 25th and 75th percentiles, with the line inside showing the median financial loss. The lines extending out, or whiskers, represent the general spread of the data. As we can see, most financial losses fall within a moderate range, and there are no extreme outliers shown beyond the whiskers. This suggests that while losses vary, most incidents stayed within a relatively predictable band without unusually massive loss events skewing the data.



This heatmap shows the number of cybersecurity incidents by attack type across each year from 2015 to 2024. The darker colors represent a higher number of incidents, while lighter colors represent fewer incidents. As we can see, phishing consistently had the highest frequency every year, making it the most common attack type across the entire period. However, starting around 2019, ransomware attacks began to rise significantly, indicating a shift in attacker strategy toward more financially motivated and damaging attacks. Other attack types like malware and insider threats remained relatively stable in frequency without major spikes. Overall, this view helps highlight both the persistence of traditional threats and the emergence of newer, more aggressive ones in recent years.

Conclusion

This project revealed several important insights into global cybersecurity threats from 2015 to

2024. Phishing remained the most common type of attack across the years, while ransomware incidents grew significantly in recent years, contributing to higher financial losses. Finance and healthcare emerged as the most frequently targeted industries, and the United Kingdom, Germany, and Brazil were the top three countries in terms of total financial loss — an unexpected finding, given the size of other global economies.

Interestingly, resolution times across different attack types were relatively consistent, suggesting that many organizations may rely on standardized response protocols. The rise in financial loss over time points to the increasing cost and complexity of cybersecurity incidents, reinforcing the need for stronger defense mechanisms and proactive strategies.

Overall, the analysis demonstrates that cyberattacks are not only frequent but also financially and operationally damaging. As the digital landscape continues to evolve, so will the nature of cybersecurity risks — making this type of data-driven insight essential for preparedness in both the public and private sectors.

Acknowledgments

I would like to thank my instructor for guiding this project and providing support throughout the process.

References

Kaggle. (2024). *Global Cybersecurity Threats Dataset (2015–2024)*. https://www.kaggle.com/datasets